

From churn model to a reliable retention service

Deployable churn predictions, governed model updates and business-aware retention decisions.

CONTROLLED CONCEPT-DRIFT EXPERIMENT

Static and adaptive branches use the same Telco customer sequence and delayed labels. Synthetic target changes are audited; business outcomes use assumed intervention uplift.

+0.016 F1

Post-shift mean: 0.569 → 0.585

+€17

Rounded cumulative simulated uplift

3 RUNS / 1 PROMOTION

Training does not force deployment

CLIENT RISK

A notebook model has no stable serving contract. Unmonitored churn risk and untargeted interventions can degrade decisions and waste budget.

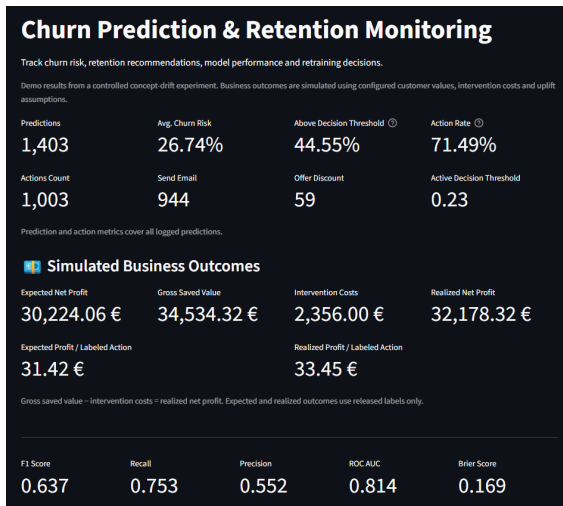
ENGINEERED RESPONSE

Validated serving bundles, guarded Champion promotion and budget-aware retention actions connect model quality to operational decisions.

DATA → FEATURES → TRAINING → REGISTRY → API → MONITORING → RETRAINING

Evidence from the working system

Local monitoring + production rollback



Business-aware monitoring

Local model quality and simulated retention outcomes; not observed campaign revenue.

Roll back prod summary

Cloud Run rollback

- Environment: `prod`
- Service: `mlops-churn-prediction-prod-api`
- Revision: `mlops-churn-prediction-prod-api-00001-b2v`
- Traffic: `100%`
- Readiness: `HTTP 200`

Verified Cloud Run rollback

Previous API revision received 100% traffic. Readiness returned HTTP 200; the follow-up semantic probe confirmed model lineage.

REPRODUCIBLE CLOUD PLATFORM

FastAPI + private MLflow on Cloud Run · Cloud SQL PostgreSQL · GCS artifacts
Terraform infrastructure · keyless GitHub Actions · renewable MLflow authentication

Cloud demo deployed, verified and torn down on 6 October 2026. Screenshots are recorded evidence.